So, we can say that at least in Python, implementing the points partitioning sub-routine by using k-d trees provides an advantage over the naïve brute-force search.

12.2.5 Final Remarks on k-means

To conclude this section on k-means, let's summarize our conclusions:

- K-means is a centroid-based hard-clustering method;
- If implemented with an auxiliary k-d tree, the running time for the algorithm is `O(m*(n*2d + d*(n+k)*log(k))`;
- K-means works well on low-to-medium-dimensional data, when cluster shapes are spherical and the number of clusters can be estimated *a priori*, but even if the dataset doesn't have a homogeneous distribution.
- K-means works poorly on high-dimensional data, and when clusters can't be approximated with hyperspheres.

Now that we have described the progenitor of all clustering algorithms, we need to fast forward 40 years before the invention of the next algorithm we are going to present.

12.3 DBSCAN

The paper introducing *DBSCAN*, in fact, was only published in 1996, presenting a novel approach to address the problem[11]; DBSCAN is an acronym for "**D**ensity-**b**ased **s**patial **c**lustering of **a**pplications with **n**oise", and the main